Bogdan Løw-Hansen
Researcher

Researcher and M.Sc. in Cybernetics and Robotics specializing in vehicle modeling, system identification, and rigid body simulation. Expertise in developing guidance, navigation, and control (GNC) algorithms for autonomous systems, including state estimation and statistical fault detection. Experience applying deep neural networks for image classification, and developing a model-based "virtual" detection sensor for UAV icing during PhD research. Appointed member of the NATO STO AVT-388 panel and selected as a 2024 Arctic Frontiers Emerging Leader.

Current Position: Researcher at the NTNU UAV Icing Lab | Engineer at UBIQ Aerospace
Email: bogdan.l.hansen@ntnu.no

Work Experience

NTNU, Trondheim Mar 2026 - Present	**Researcher at the Dept. of Engineering Cybernetics** Developing ice protection solutions for green aviation under the Zero Ice Shield research project.
UBIQ Aerospace, Trondheim Mar 2023 - Present	**Modeling and Simulation Engineer** Delivering experimental and simulation-based analyses to support engineering operations in UAV icing conditions.
DTU Electro, Denmark Jan 2025 - Jul 2025	**Research Stay at DTU in Denmark, visiting the CREA group** Conducted statistical model-based real-time detection of propeller icing on UAVs.
NTNU, Trondheim Aug 2021 - Mar 2026	**Ph.D. Candidate at the Dept. of Engineering Cybernetics** Researched and developed autonomous detection algorithms for in-flight icing on small fixed-wing UAVs.
NTNU, Trondheim Aug 2022 - Jan 2026	**Scientific Assistant** TTK4190 - Guidance, Navigation and Control of Vehicles Managed course assignments, supervised students, and graded exams.
NTNU, Trondheim Aug 2020 - Apr 2021	**Student Assistant** Assisted in TTK4260 Multivariate analysis and Machine learning methods and TTK4190.
Aker BioMarine, Oslo Jun 2020 - Aug 2020	**Summer Intern** • Implemented a convolutional neural network (CNN) using PyTorch for plankton image classification. • Trained models remotely on Google Cloud Platform, outperforming baseline manual feature engineering methods. Results were shared with the Institute of Marine Research.
Ingeniarius, Portugal Jul 2019 - Sep 2019	**RobotCraft 2019, Internship/Summer-Course** • Designed software for a wheeled robot to autonomously map and solve a maze-navigation task. • Implemented the navigation algorithm fusing 2D Lidar scans and motor encoders using ROS and the A* algorithm.

Professional Affiliations & Awards

NATO STO Mar 2026 - Present	**Member of the Applied Vehicle Technology (AVT-388) Group** Operation of Unmanned Aerial Vehicles (UAVs) in Icing Environments.
Emerging Leader 2024	Issued by Arctic Frontiers. Selected for a 1-week mentorship program for young global leaders addressing Arctic challenges.

Publications

(2026) System Identification of Leading-Edge Rime-Ice Effects on a Small Fixed-Wing Drone *(Under review)*
(2026) Real-time Detection of Propeller Icing for Electric Fixed-Wing UAVs Using Estimated Propeller Torque *(To be submitted)*
(2025) Modeling and Identification of a Small Fixed-Wing UAV Using Estimated Aerodynamic Angles. *CEAS Aeronautical Journal*
(2024) Time-Domain System Identification and Validation of Small Fixed-Wing UAV Dynamics. *AIAA Aviation Forum*
(2023) UAV Icing: A Survey of Recent Developments in Ice Detection Methods. *IFAC WC 2023*
(2023) Identification of an Electric UAV Propulsion System in Icing Conditions. *SAE Technical Paper Series*
(2022) UAV Icing: Ice Shedding Detection Methods for an Electrothermal De-Icing System. *AIAA Aviation Forum*
Müller et al., (2023) UAV icing: Development of an ice protection system for the propeller of a small UAV. *Cold Regions Science*

Education

NTNU, Trondheim
Aug 2015 - Jul 2021

M.Sc. in Cybernetics and Robotics
Master's thesis: Electrothermal Ice-Shedding Detection for UAVs.
Specialization Project: Lidar-based Positioning for Docking of Autonomous Ships.

KTH, Sweden
Aug 2018 - Jun 2019

Exchange Student in Stockholm
Coursework focus: Signal processing, non-linear control, MIMO-control, and embedded control systems.

Student Organisations

NTNU Ascend
Sep 2019 - Aug 2020

Aerial Robotics, Team 2020
Implemented lidar-based obstacle avoidance and path-following control.

KTH FS
Sep 2018 - Jun 2019

Formula Student, Team 2019
Worked with vehicle component tuning and sensor data analysis.

Computer Skills

Languages:	MATLAB, Python, C++, Bash/Shell.
Frameworks & Tools:	Simulink, PyTorch, ROS, Docker, Git (GitLab/GitHub), Linux (Ubuntu), Google Cloud Platform (GCP).
Techniques:	Modeling and system identification, autonomous vehicle guidance navigation and control (GNC), state estimation/Kalman filtering, data analysis and visualization, convolutional neural networks (CNN).

Other Work Experience

Forsvaret
Sep 2014 - Aug 2015

Military service in Norwegian Navy (Haakonsvern)
Served aboard KNM Rauma and KNM Hitra.

Languages & Certificates

Norwegian:	Fluent
English:	Professional Proficiency
Ukrainian:	Conversant

2022	Driving license, class - A and B
2012	Forklift license, class - T4